

Answer key

LING 201 Midterm Review

⚠ Questions on this handout might not be exactly what will appear on the final. ⚠

⚠ Please look at past slides and exercises for review as well. ⚠

1. Introduction to Linguistics

- **Linguistics** is the *scientific* study of language.
- Linguists use the scientific method to uncover the *competence* or knowledge of language users.
- While there are both prescriptive and descriptive rules relating to language, linguists are interested in uncovering the *descriptive rules* of a language.
 - **Descriptive Rules:** Rules that describe how people use their language. These rules are often learned and used tacitly.
 - **Prescriptive Rules:** Rules that tell people how they should speak (or write). Usually require some effort to learn. Are often broken.
- Often times the descriptive rules are *tacit*, hence speakers are not aware that they are following them.
- We can use the scientific method to uncover rules of languages we ourselves speak or languages we are unfamiliar with.

2. Morphology

- **Morpheme** is the *smallest* linguistic unit that carries meaning or grammatical function.
- **Complex** word: Word which contains more than one morpheme.
- **Simple** word: Word which consists of a single morpheme.
- **Free** morphemes can stand alone as a word. e.g. *dog, cat, house, . . .*
- **Bound** morphemes cannot stand alone. e.g. *-er, -s, cran-, . . .*
 - Affixes: prefixes, suffixes, infixes, circumfixes
 - Bound roots: *cran-, rasp-*
- **Root:** the 'base' morpheme to which affixes attach. These contribute the 'basic' meaning.
- **Derivational** morphemes change meaning, creating new words.
 - Often, but not always, they also change the part of speech.
 - *clever-ness, enjoy-ment, lion-ess, deriv-ation-al*
- **Inflectional** morphemes provide grammatical information, and indicate 'agreement' or 'conjugation' in some contexts.
 - They do not change part of speech of the base.
 - Tense (*-ed*), aspect (*-ing*), number (*-s*), person (*-s*), noun class, case

- Morpheme boundaries *do not necessarily correspond* with other linguistics divisions like syllable boundaries.
 - A single morpheme in one language can be translated into more than one morpheme (or even multiple words) in another language.
 - Just because a morpheme corresponds to a word in English does not mean it is necessarily a word in the given language.
- **Foreign morphemic analysis** - Swahili (from Assignment 1)

The general procedure for handling morphemic analysis problems:

- 1) First, break down the English translations into morphemes.
 e.g., 'I see them' = I + present tense + see + them.
 'You will see me' = you + will + see + me.
 'I saw you' = I + past + see + you.
- 2) Then find two translations that are either:
 - the exact same except for one morpheme that's different – for example, 'I speak' and 'you speak'. OR
 - completely different except for one morpheme that's the same – for example, 'she speaks' and 'she sees me'.
- 3) Finally, compare the English translation and the foreign word.
 - If you found two translations that are the same except for one morpheme, then whatever is different between the foreign words = the one morpheme that's different between the translations.
 - If you found two translations that are completely different except for one morpheme, then what is the same between the foreign words = the one morpheme that's the same between the translations.
 - Note that the order of morpheme in that language may be different from that in English.

Exercise #1: Revisiting Swahili

Consider the following examples from Swahili of Eastern Congo and answer the questions below.

- | | | | | | |
|------|-----------|---------------|-------|--------------|--------------------|
| i. | Ninasema. | ‘I speak.’ | vi. | Niliona. | ‘I saw.’ |
| ii. | Wunasema. | ‘You speak.’ | vii. | Ninawaona. | ‘I see them.’ |
| iii. | Anasema. | ‘She speaks.’ | viii. | Niliwuona. | ‘I saw you.’ |
| iv. | Wanasema. | ‘They speak.’ | ix. | Ananiona. | ‘She sees me.’ |
| v. | Ninaona. | ‘I see.’ | x. | Wutakaniona. | ‘You will see me.’ |

- (1) Try to match the meanings below with a morpheme in the Swahili data. Hints: The morphemes for subject and object pronouns are the same (e.g., the morpheme for ‘I’ is the same as the morpheme for ‘me’).

speak	<u>sema</u>	I/me	<u>ni</u>
see	<u>ona</u>	they/them	<u>wa</u>
present	<u>na</u>	you	<u>wu</u>
past	<u>li</u>	will	<u>taka</u>

3. Syntax

3.1. Parts of speech/syntactic categories

- Noun (N):
 - can combine with plural morpheme *-s*;
 - often ends with *-ion/-ment/...*;
 - can occur after a determiner, ...
- Verb (V):
 - can combine with past morpheme *-d*, has combined with *-ize*, ...
 - can follow an auxiliary (*does*), modal (*might*), or negation (*not*), ...
- Adjective (Adj):
 - can combine with *un-*, has combined with *-able*, ...
 - can occur between a determiner and a noun, ...
- Adverb (Adv):
 - appear with *-ly*, ...
 - can’t appear between a determiner and a noun, ...

3.2. Open vs. closed parts of speech

- **Open:** N, V, Adj, Adv
- **Closed:** D, P, Conj, C, T, Neg, ...

3.3. Lexical vs. functional parts of speech

- **Lexical** parts of speech (N, V, Adj, Adv) provide the *content* of the sentence.
- **Functional** parts of speech (D, P, Conj, C, T, Neg) provide the *grammatical* information.

3.4. Examples of some syntactic categories

- Determiner (D): *a/an, the, every, some, no, most, ...*
- Preposition (P): *over, under, in, near, on, to, with, of, . . .*
- Conjunction (Conj): *and, or, neither-nor, either-or, . . .*
- Complementizers (C): *that, whether, if, . . .*
- Tense (T):
 - Auxiliary: *has/have/had, am/is/are/was/were, do/does/did*
 - Modals: *will, would, shall, should, can, could, may, might, must*
 - Non-finite tense marker: *to*
- Negation (Neg): *not*

3.5. Constituency tests

- A group of words that function as a unit is called a **constituent**.
- Constituency test applies to a portion of a sentence and identify whether groups of words act together as a unit or not.
- If the constituency test doesn't apply, we can do the same way but using * (ungrammatical) notation to show that the test doesn't work.
- The important point is that when applying these tests to a group of words, the sentence *still needs to make sense*.

- 1) **Replacement** test: If a string of words can be replaced with by a pronoun (*he/she/they*), an adverb (*here/there*) or a do verb (*did/did so*), then that string of words is a constituent.

Example:

- The students from New Brunswick stopped by. → They stopped by.
- Mary went to school. → Mary went there.
- Mary went to school. → Mary did so.

2) **Fragment** test: If a string of words can serve as the answer to some *wh*-question, then it is a constituent.

○ Example:

Q: Who stopped by?

A: The students from New Brunswick.

3) **Movement** tests: If a string of words can be moved, then it is a constituent.

i. Topicalization:

Example: I saw **a sheep**. → **A sheep**, I saw.

ii. Passivation:

Example: I saw **a sheep**. → **A sheep** was seen by me.

iii. Clefting:

Example: I saw **a sheep**. → It was **a sheep** that I saw.

iv. Pseudo-clefting:

Example: I saw **a sheep**. → What I saw was **a sheep**.

Exercise #3: Constituency test

The students loved their syntax assignment.

- Is this a constituent? **Yes**
- Replacement test: **The students loved it.**
- Fragment test: **Q: What did the students love? A: Their syntax assignment.**
- Movement test: **It was their syntax assignment that the students loved.**

The students loved their syntax assignment.

- Is this a constituent? **No**
- Replacement test: ***The students loved it assignment.**
- Fragment test: **Q: What did the students love? A: *Their syntax.**
- Movement test: ***Their syntax, the students loved assignment.**

3.6. Phrase structure rules

CP → (C) TP

TP → {NP, CP} T VP

NP → (D) (AdjP+) N (PP+) (CP)

PP → P NP

AdjP → (AdvP) Adj

AdvP → (AdvP) Adv

VP → (AdvP+) V (NP) ({NP, CP}) (AdvP+) (PP+) (AdvP+)

⚠ **Tips for drawing trees:** ⚠

1. Every sentence is a TP, which means you should have a TP node as the top-most node, **even if there is no visible T**.
 - T can be overt as 'will', or it can also be covert when the tense of the sentence is past tense.
2. Be careful whether a PP is grouped under an NP or a VP (think about the meaning of the sentence to help decide).
3. When you have an Adj, make sure it is first connected to an AdjP node before you connect it to anything else.

In other words, you should have:

AdjP
|
Adj
Smart

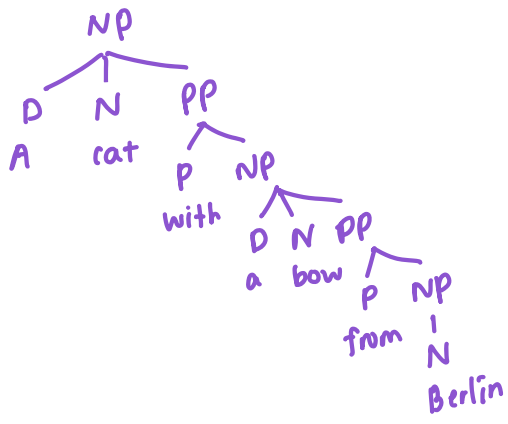
And you should not have only:

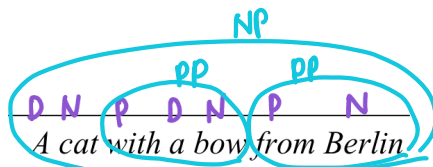
AdjP
|
Smart

You also should not have only:

Adj
Smart

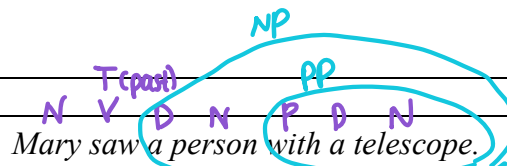
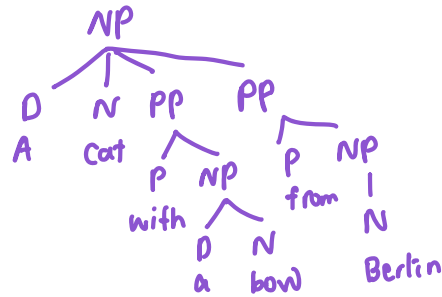
Exercise #4: Structural ambiguity

	NP D N P D N P N <i>A cat with a bow from Berlin</i>
Interpretation 1:	A bow is from Berlin PP
Syntactic tree:	 <pre> graph TD NP1[NP] --- D1[D] NP1 --- N1[N] NP1 --- PP1[PP] D1 --- A[A] N1 --- cat[cat] PP1 --- P1[P] PP1 --- NP2[NP] P1 --- with[with] NP2 --- D2[D] NP2 --- N2[N] NP2 --- PP2[PP] D2 --- a[a] N2 --- bow[bow] PP2 --- P2[P] PP2 --- NP3[NP] P2 --- from[from] NP3 --- N3[N] N3 --- Berlin[Berlin] </pre>



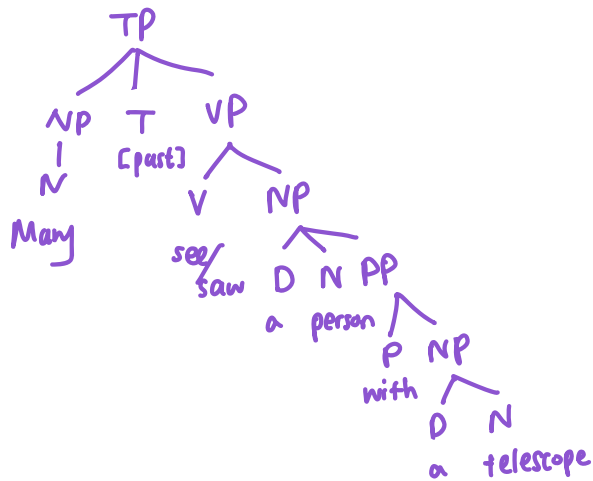
Interpretation 2: A cat is from Berlin

Syntactic tree:



Interpretation 1: A person has a telescope

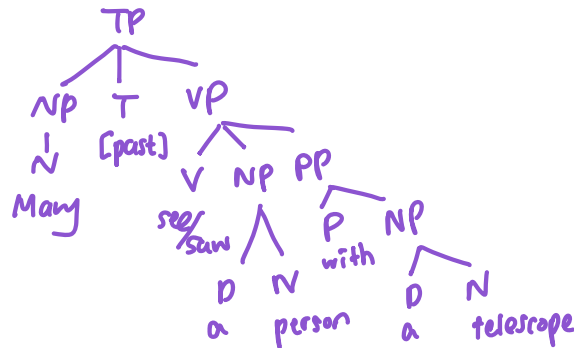
Syntactic tree:



$\overline{N} \overline{V} \overline{D} \overline{N} \overline{P} \overline{D} \overline{N}$
 Mary saw a person with a telescope

Interpretation 2: Mary has a telescope

Syntactic tree:



4. Semantics

4.1. Set theory

- **Definition:** A **set** is an abstract collection of *distinct* objects.
- **Important:** objects in a set must be *distinct*. Two of the same objects in a set counts as one – e.g. $\{a, a, b\} = \{a, b\}$.
- Note that sets can also be objects in a set – a set of sets.

4.2. Some important notions about sets

- **Membership:** We write $a \in S$ if a is a *member* of the set S , and $a \notin S$ otherwise.
e.g., Suppose $S = \{a, b, c\}$. Then $a \in S$, and $d \notin S$.
- **Identity:** set A and set B are *identical/equal* iff they have exactly the same members, written as $A = B$. If A and b are not identical/equal, we write $A \neq B$.
e.g., Let $A = \{a, b, c\}$, $B = \{a, b, c\}$. Then $A = B$.
- **Subset:** set A is a *subset* of set B iff every element of A is also an element of B , written as $A \subseteq B$. If A is not a subset of B , we write $A \not\subseteq B$.
e.g., Let $A = \{a, b, c\}$, $B = \{a, b, c, d, e\}$. Then $A \subseteq B$.
- **Superset:** set B is a *superset* of set A iff A is a subset of B , written as $A \subseteq B$. If B is not a superset of A , we write $A \not\supseteq B$.
e.g., Let $A = \{a, b, c\}$, $B = \{a, b, c, d, e\}$. Then $B \supseteq A$.

Note: there are also two special sets:

- **The empty set:** The set with no members. Written as either $\emptyset$ or nothing $\{ \}$.
- **The universe of discourse:** The set containing everything. Written as U .

4.3. Operations we can do on sets

- **Union:** written as $A \cup B$. The union of A and B simply adds the elements in each set together.
e.g., Let $A = \{a, b, c\}$, $B = \{b, c, d\}$. Then $A \cup B = \{a, b, c, d\}$.
- **Intersection:** written as $A \cap B$. The intersection of A and B singles out only the elements that are found in both sets.
e.g., Let $A = \{a, b, c\}$, $B = \{b, c, d\}$. Then $A \cap B = \{b, c\}$.
- **Difference:** written as $A - B$. $A - B$ picks out elements that are only in A and not in B.
e.g., Let $A = \{a, b, c\}$, $B = \{b, c, d\}$. Then $A - B = \{a\}$.
- **Complement:** The complement of set A is written as A' . The complement of A is the difference between the set containing everything and A.
e.g., Let $A = \{a, b, c\}$. Then A' is the set that contains anything but a, b, and c.

4.4. The extensions/meanings of ...

- **Sentence:** truth value (1 for true, 0 for false).
- **Name:** a single, specific individual/entity. Lower case letters.
e.g., $[[Bart]] = b$, $[[Lisa]] = l$, $[[Maggie]] = m$.
- **Common noun:** a set of individuals.
e.g., $[[student]] = \{b, l\}$
- **Adjective:** a set of individuals.
e.g., Suppose Bart and Lisa are sleepy; then $[[sleepy]] = \{b, l\}$
- **Verb Phrase:** a set of individuals
e.g., Suppose Lisa and Maggie cried; then $[[cried]] = \{l, m\}$
e.g., Suppose Lisa and Bark go to school; then $[[go to school]] = \{b, l\}$

4.5. Semantic composition

- **Simple sentences (noun phrase + verb phrase):**

The meaning of a simple sentence in the form of 'NP VP': $\llbracket \text{NP VP} \rrbracket = \llbracket \text{NP} \rrbracket \in \llbracket \text{VP} \rrbracket$

e.g., Suppose Maggie cried, Lisa is a student, Bart is a semanticist;

then $\llbracket \text{Maggie cried} \rrbracket = \llbracket \text{Maggie} \rrbracket \in \llbracket \text{cried} \rrbracket$
 $= m \in \{l, m\}$
 $= 1 \text{ (true)}$

then $\llbracket \text{Lisa is a student} \rrbracket = \llbracket \text{Lisa} \rrbracket \in \llbracket \text{is a student} \rrbracket$
 $= l \in \{b, l\}$
 $= 1 \text{ (true)}$

then $\llbracket \text{Bart is a semanticist} \rrbracket = \llbracket \text{Bart} \rrbracket \in \llbracket \text{is a semanticist} \rrbracket$
 $= b \in \{ \}$
 $= 0 \text{ (false)}$

Exercises #3: Semantic composition

Assume the toy universe given below:

Anton, Berta, Gusti, Hanna, Josef, Mariam, Rosi, and Susi are Canadians.

Anton, Berta, Chris, Hanna, Leo, Mariam, Susi are Zoe are skiers.

David, Ian, Mariam, Susi, Tim, and Zoe are Rutgers students.

Give the meaning of the following items.

$\llbracket \text{Anton} \rrbracket = a$	$\llbracket \text{Josef} \rrbracket = j$
$\llbracket \text{Berta} \rrbracket = b$	$\llbracket \text{Leo} \rrbracket = l$
$\llbracket \text{Chris} \rrbracket = c$	$\llbracket \text{Mariam} \rrbracket = m$
$\llbracket \text{David} \rrbracket = d$	$\llbracket \text{Susi} \rrbracket = s$
$\llbracket \text{Gusti} \rrbracket = g$	$\llbracket \text{Rosi} \rrbracket = r$
$\llbracket \text{Hanna} \rrbracket = h$	$\llbracket \text{Tim} \rrbracket = t$
$\llbracket \text{Ian} \rrbracket = i$	$\llbracket \text{Zoe} \rrbracket = z$

$\llbracket \text{Canadians} \rrbracket = \{ a, b, g, h, j, m, r, s \}$
$\llbracket \text{skiers} \rrbracket = \{ a, b, c, h, l, m, s, z \}$

$$\llbracket \text{Rutgers students} \rrbracket = \{d, i, m, s, t, z\}$$

$$\begin{aligned}\llbracket \text{Anton is a Canadian} \rrbracket &= \llbracket \text{Anton} \rrbracket \in \llbracket \text{is a Canadian} \rrbracket \\ &= \underline{a} \in \{ \underline{a}, b, g, h, j, m, r, s \} \\ &= 1 \text{ (true)}\end{aligned}$$

$$\begin{aligned}\llbracket \text{Chris is a skier} \rrbracket &= \llbracket \text{Chris} \rrbracket \in \llbracket \text{is a skier} \rrbracket \\ &= \underline{c} \in \{ a, b, \underline{c}, h, l, m, s, z \} \\ &= 1 \text{ (true)}\end{aligned}$$

$$\begin{aligned}\llbracket \text{Leo is a Rutgers student} \rrbracket &= \llbracket \text{Leo} \rrbracket \in \llbracket \text{is a Rutgers student} \rrbracket \\ &= l \in \{ d, i, m, s, t, z \} \\ &= 0 \text{ (false)}\end{aligned}$$